Electrical Engineering Portfolio

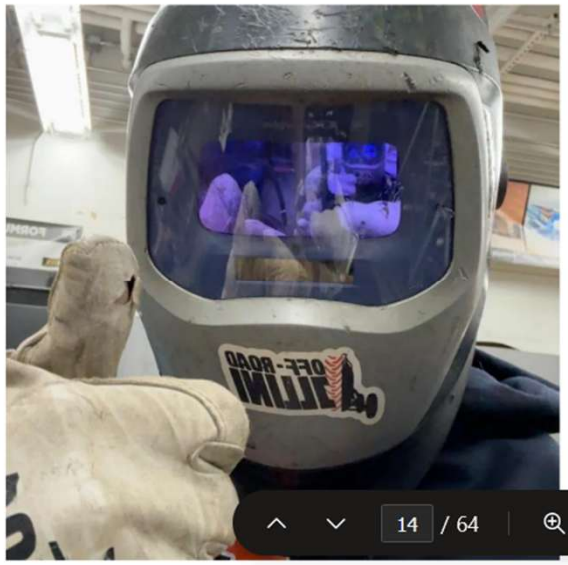
Daniel Salkin

Research & Development – Daniel Salkin

Illini Electric Motorsports

- From Los Angeles, CA
- Sophomore
- Developed the Dash Board PCB last year
- Firmware Dev. @ John Deere
- Amateur Baker







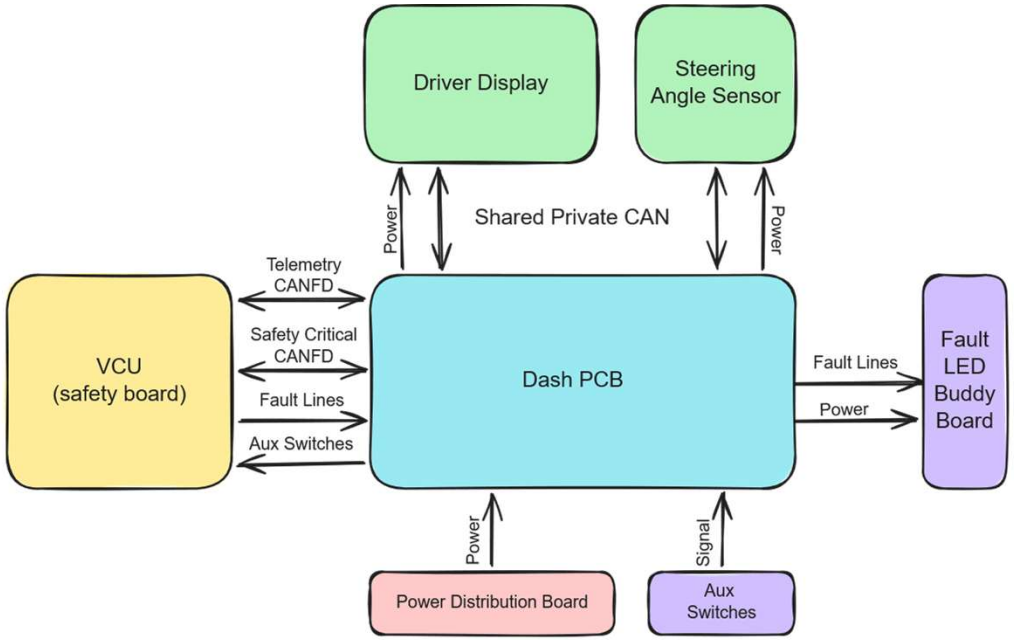


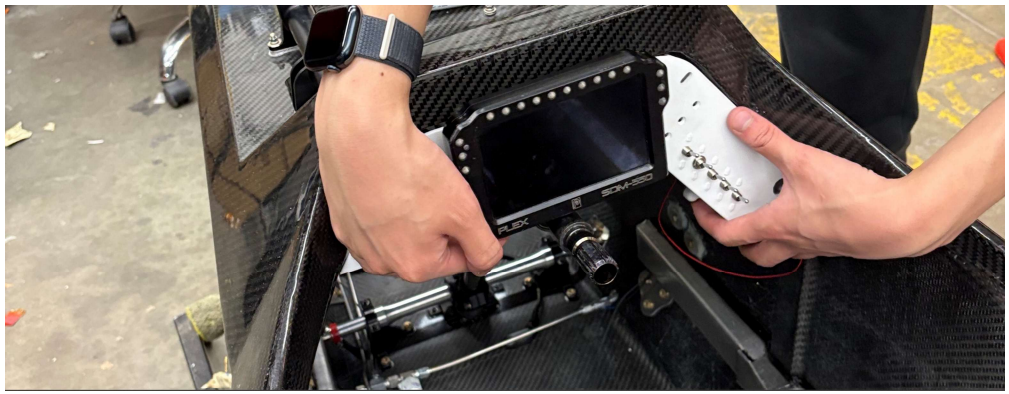




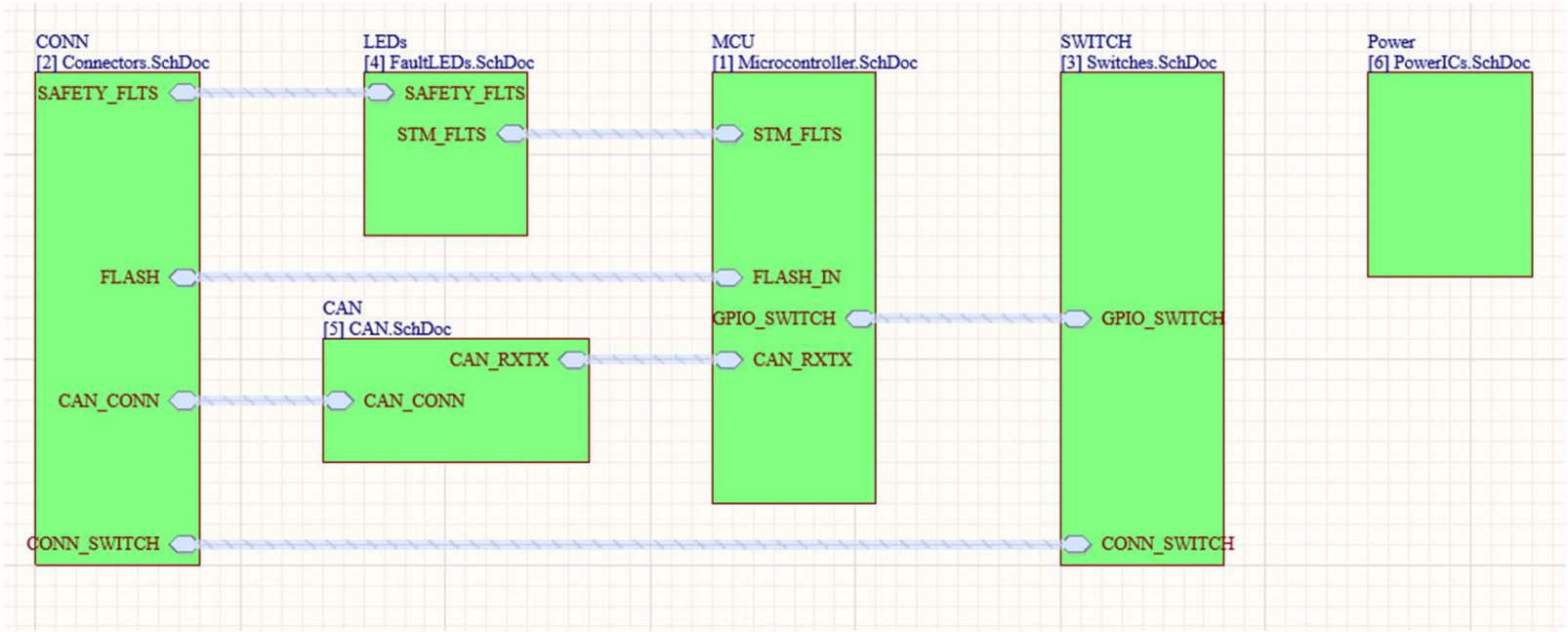
Custom Dash PCB

- Provides power to and communicates with the driver display and Steering angle Sensor
- Acts as a CAN bridge between the CAN 2.0 display and CANFD telemetry bus
- Handles rules critical fault indication redundant to driver display
- Handles driver I/O for Car Enable, Torque Vectoring, Cooling, and Regen

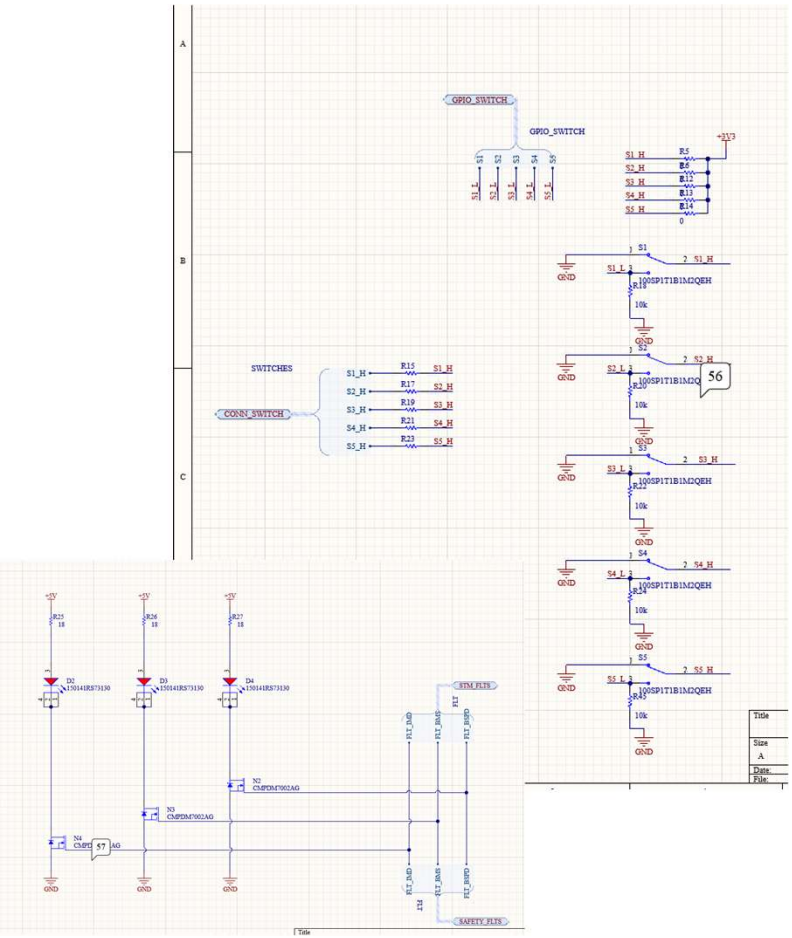


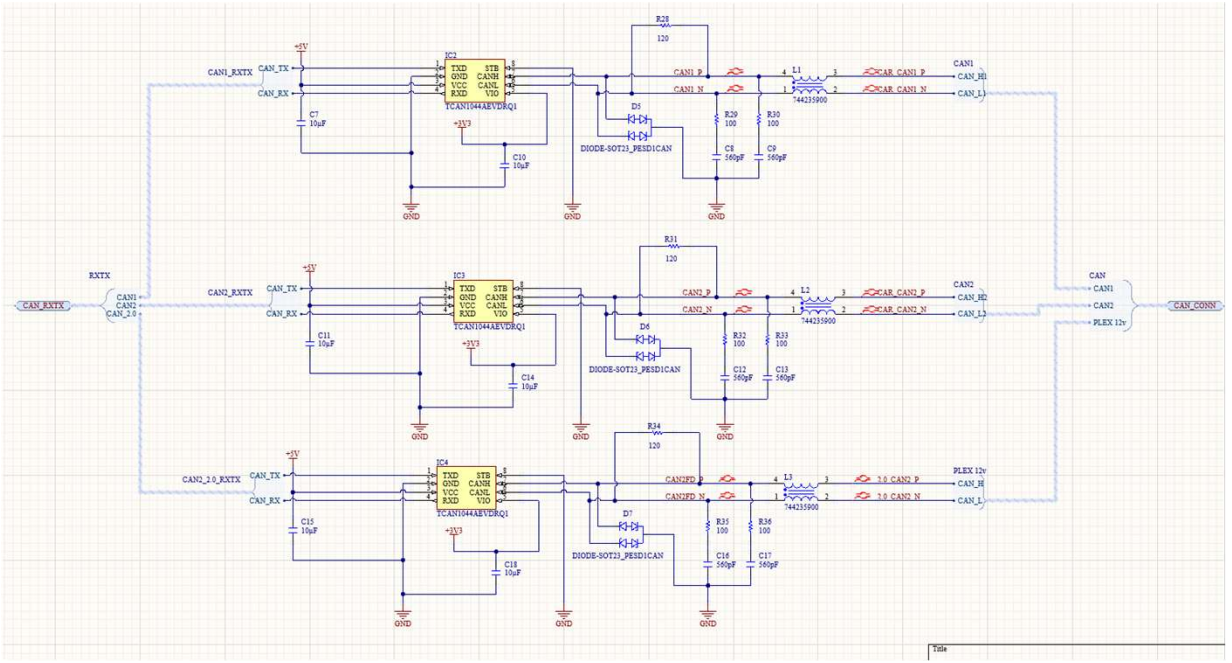


Top Level Schematic

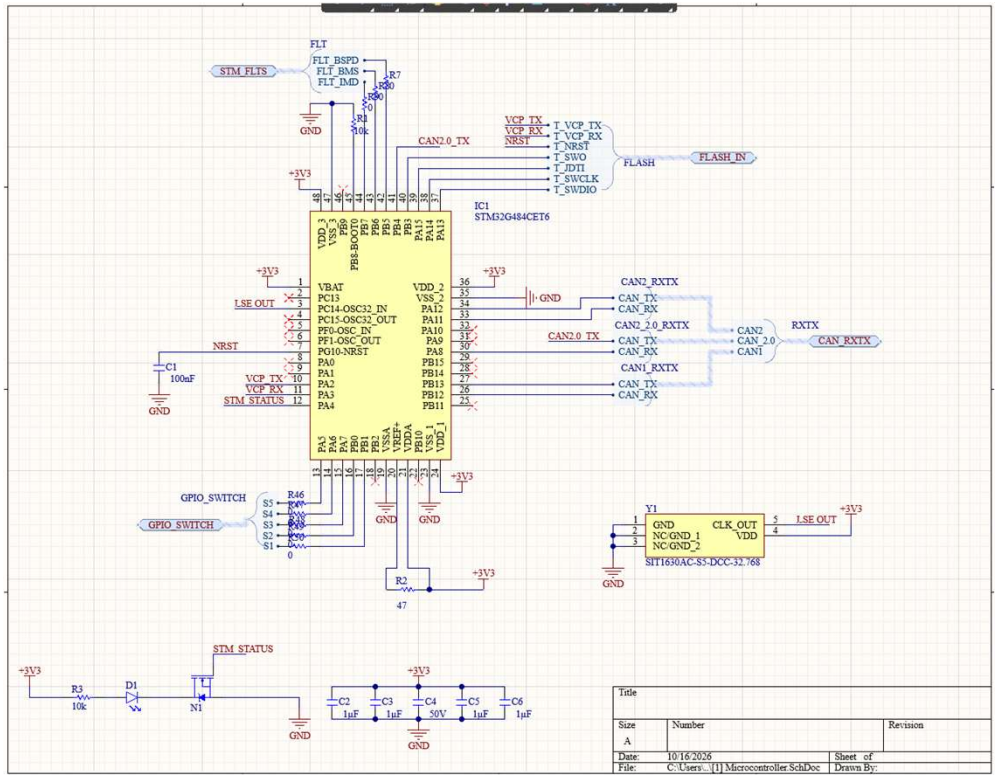


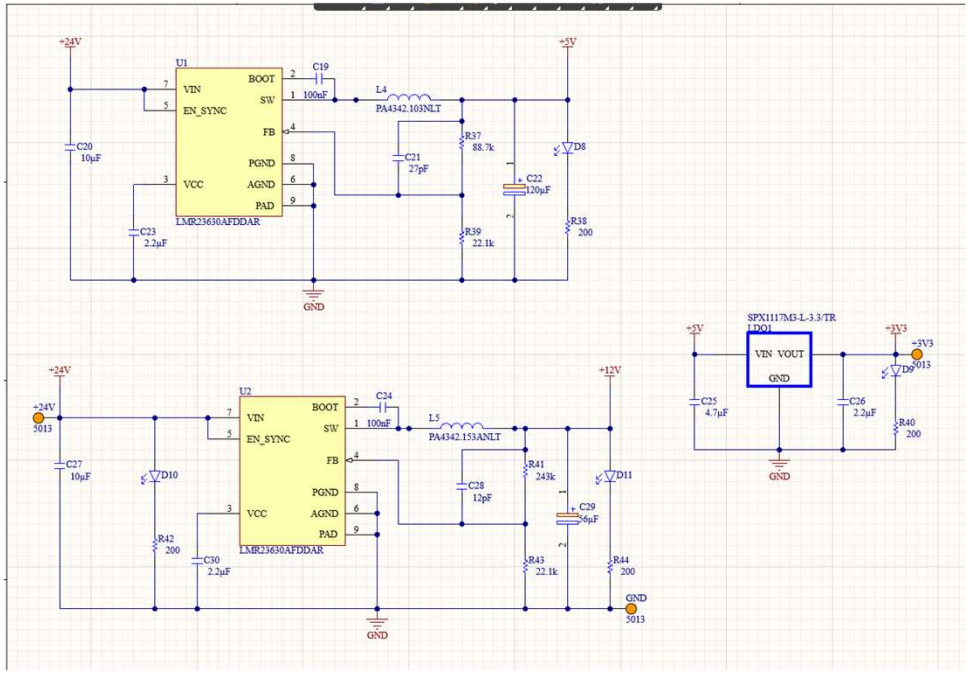
CAN Transcievers, Faults, Switches





Microcontroller & Power ICs





CAN Architecture Redesign

- Illini Electric Motorsports experienced data corruption and frame drops over their telemetry and data critical busses
- I was tasked with finding the root cause of these issues and implement solutions
- After implementing these changes we were able to prove functionality of 4Mbps CANFD across car

Harness & Architecture Improvements

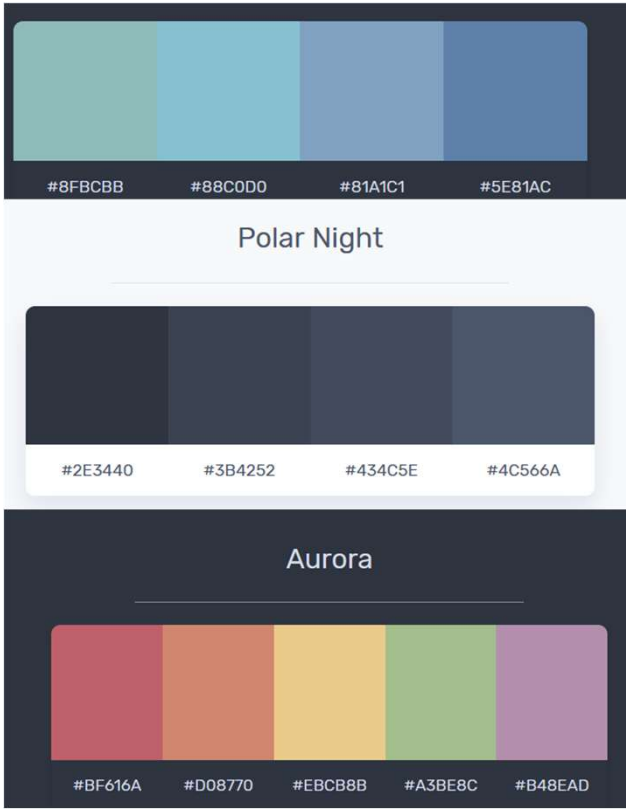
- Limit stub lengths to <20cm based off of 1.72 critical length @1Mbps for
  - Stubs of 1m must be treated as transmission lines

CAN Transceiver Circuit Redesign

- Illini Electric Motorsports experienced data corruption and frame drops over their telemetry and data critical busses
- I was tasked with finding the root cause of these issues and implement solutions
- After implementing these changes we were able to prove functionality of 4Mbps CANFD across car

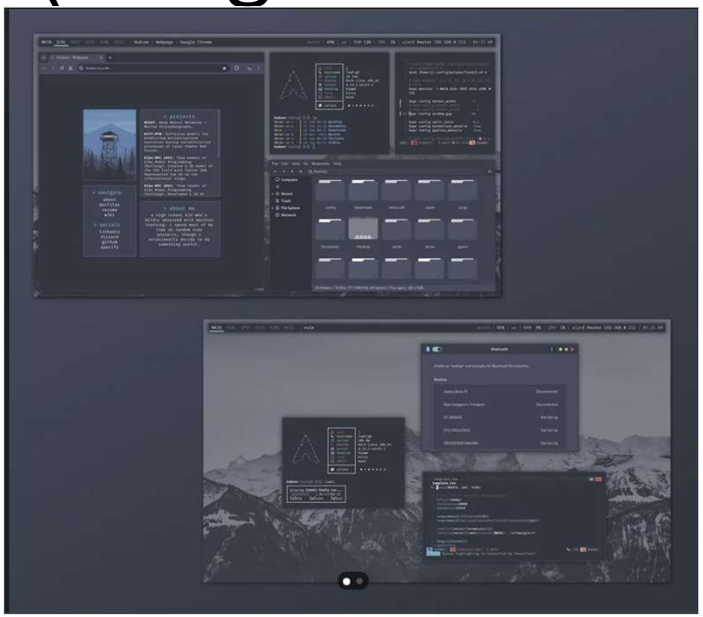
design

- Color palette: Nord (polar night, aurora, frost)
- Slightly rounded geometric rectangles
- Sidebar with home projects experience contact
- Sans serif(like Cascadia code)
- Make it readable when you scroll



Reference images (designs I think are tuff)







	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Year:	Freshman		Sophmore		Junior		Senior		Total			
2	Semester:	Fall	Spring	Fall	Spring	Fall	Spring	Fall	Spring				
3		ECE 120 (4)	ECE 110 (3)	PHYS 213 (2)	ECE 411 (4)	ECE 391 (4)	ECE 470 (4)	ECE 486 (4)	ECE 445 (4)		Free Elective		
4		ENG 100 (1)	ECE 220 (4)	PHYS 214 (2)	ARTJ 302 (3)	ECE 310 (3)	ECE 340 (3)	ECE 350 (3)	ECE 464(3)		CE Technical		
5		MATH 441 (3)	MATH 213 (3)	ECE 210 (4)	ECE 329 (3)	ECE 311 (1)	ECE 425 (3)	ECE 483 (3)	ECE 462 (3)		Advanced Computing		
6		PHYS 211 (4)	PHYS 212 (4)	CS 225 (4)	ECE 330(3)	ECE 342/343(3)	ME199 (1)	FSHN 343 (2)	MACS 100 (3)		Core		
7		ggis 254 (3)	MATH 257 (3)	ECE 385 (3)	buh (0)	ECE 343(1)	KIN 104(1)	buh (0)	buh (0)		Gened		Hardware systems cou
8	Hours:	15	18	15	13	12	12	12	13	110			
9			ECE 145 (1)		maybe ece 340			**ece 453**					
10						IE 420/421		ECE 427					
11		GE Needs:	Humanities&Arts (6)									TAM 211 ->	TAM 212 ->
12			Western/Comparative										AE 352 ->
13			Non-Western										
14											CS 225->	CS 461	Computer security I
15												CS 341	

Refer to this: https://jonathan-tao.github.io/

https://www.reddit.com/r/unixporn/comments/1iv5d2v/bspwm_minimal_nord_rice/

https://www.reddit.com/r/hyprland/comments/1iok5kn/hyprland_rate_my_first_rice_on_hyprland/